

Summation of Postsynaptic Potentials

The interplay between multiple excitatory and inhibitory inputs is the essence of integration in the nervous system. The cell body and dendrites of a given postsynaptic neuron may receive inputs from chemical synapses formed with hundreds or even thousands of synaptic terminals (see Figure 48.2). How do so many synapses contribute to information transfer?

The input from an individual synapse is typically insufficient to trigger a response in a postsynaptic neuron. To see why, consider an EPSP arising at a single synapse. As a graded potential, the EPSP becomes smaller as it spreads from the synapse. Therefore, by the time a particular EPSP reaches the axon hillock, it is usually too small to trigger an action potential (Figure 48.17a).

On some occasions, individual postsynaptic potentials combine to produce a larger postsynaptic potential, a process called **summation**. For instance, two EPSPs may occur at a single synapse in rapid succession. If the second EPSP arises before the postsynaptic membrane potential returns to its resting value, the EPSPs add together through *temporal summation*. If the summed postsynaptic potentials depolarize the membrane at the axon hillock to threshold, the result is an action potential (Figure 48.17b). Summation can also involve multiple synapses on the same postsynaptic

neuron. If such synapses are active at the same time, the resulting EPSPs can add together through *spatial summation* (Figure 48.17c).

Summation applies as well to IPSPs: Two or more IPSPs occurring nearly simultaneously at synapses in the same region or in rapid succession at the same synapse have a larger hyperpolarizing effect than a single IPSP. Through summation, an IPSP can also counter the effect of an EPSP (Figure 48.17d).

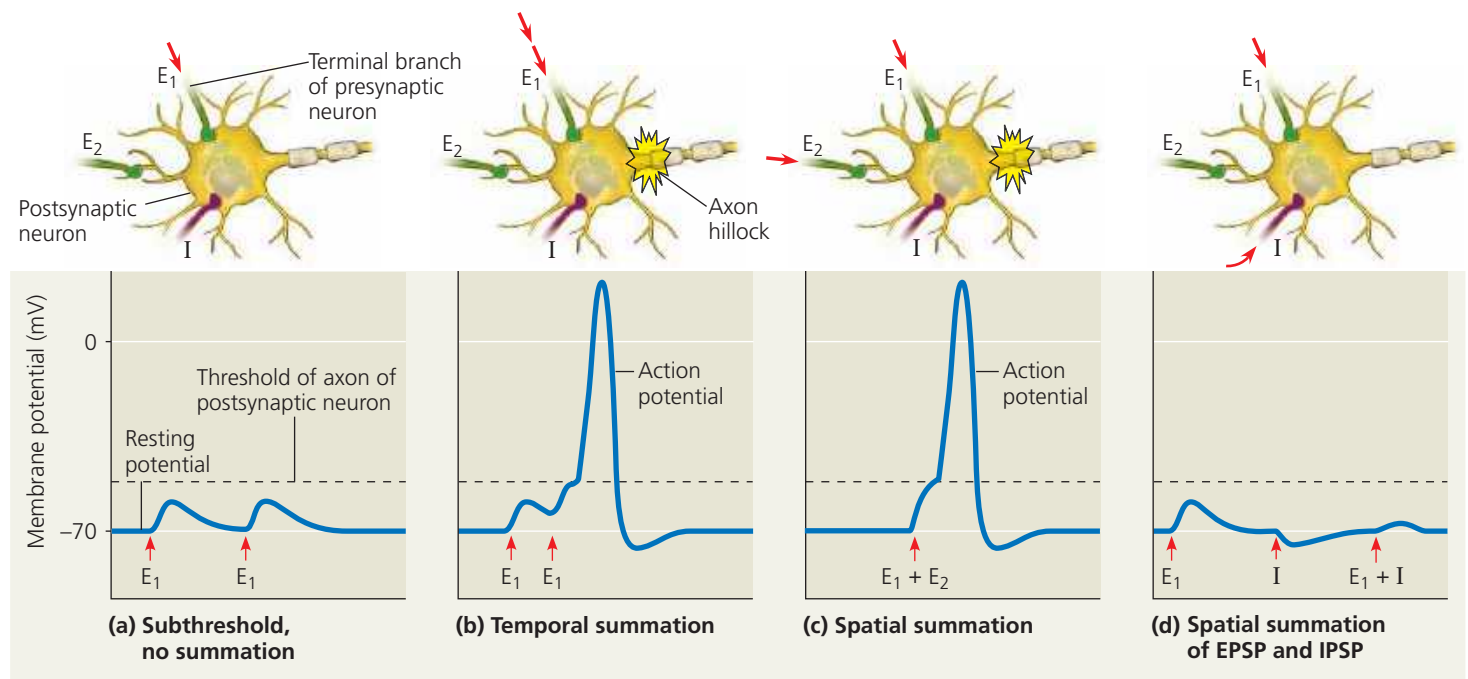
The axon hillock is the neuron's integrating center, the region where the membrane potential at any instant represents the summed effect of all EPSPs and IPSPs. Whenever the membrane potential at the axon hillock reaches threshold, an action potential is generated and travels along the axon to its synaptic terminals. After the refractory period, the neuron may produce another action potential, provided the membrane potential at the axon hillock once again reaches threshold.

Termination of Neurotransmitter Signaling

After a response is triggered, the chemical synapse returns to its resting state. How does this happen? The key step is clearing the neurotransmitter molecules from the synaptic cleft. Some neurotransmitters are inactivated by enzymatic

▼ **Figure 48.17 Summation of postsynaptic potentials.** These graphs trace changes in the membrane potential at a postsynaptic neuron's axon hillock. The red arrows indicate times when postsynaptic potentials occur at two excitatory synapses (E_1 and E_2 , green in the diagrams above the graphs) and at one inhibitory synapse (I , purple). Like most EPSPs, those produced at E_1 or E_2 do not reach the threshold at the axon hillock without summation.

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Animation: Action Potentials



VISUAL SKILLS Using these drawings, propose an argument for all summation being in some sense temporal.